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Department of Computer Science and Engineering

Assignment-1 (Mid-term), Autumn-2025

Course: **CSE-3521** (**Computer Architecture)**

Marks: **10**

[Answer all ***three*** questions; Figures in the right hand margin indicate Marks]

Q1.	Consider the following C statements with five integer variables `a, b, c, d, e`:
	
	`a = b + c;`
	`d = a - e;`
	
	Translate the equations in MIPS assembly instructions. Calculate the Machine code. Assume, a, b, c, d, e will be \$t0, \$t1, \$s0, \$s1, \$s2.
	
	Explain:
	
	1. Suppose the variables `a` and `e` are stored in the same memory location by mistake. Explain what happens when these instructions execute.
	2. Write the expected values of `a` and `d` after execution in this scenario.
		5	**CO1**
			CO2

Q2.	The *first version of multiplication hardware* multiplies two integers `x` and `y` by performing repeated addition: adding `x` to an accumulator `y` times. A student decides to reuse the accumulator memory for both `x` and `z` to save hardware registers. Draw the flowchart. Explain: 1. Without computing exact numbers, explain why the result of `z = x * y` may be incorrect in this scenario. 2. Describe a step-by-step sequence of additions showing how the accumulator changes if `x = 4` and `y = 3`. 3. Suggest a creative fix that allows the hardware to still use minimal registers but always produces the correct result.	5	**CO2** **CO3**